

A Very Good Method for Bayes-Adaptive Deep RL via Meta-Learning(variBAD)

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Introduction

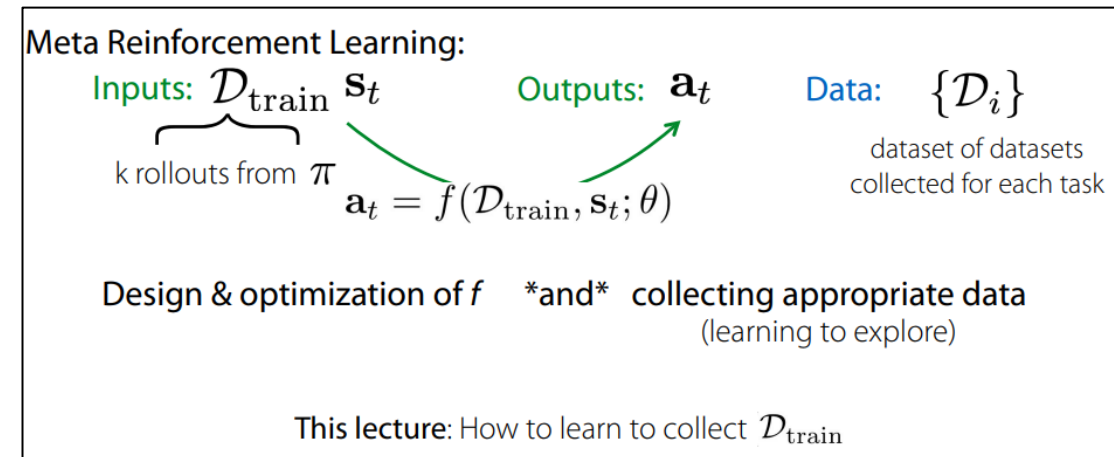
Meta Reinforcement Learning

● Problem setting of reinforcement learning, meta-learning

- Reinforcement learning
 - Given certain MDP, learn a policy π that maximize the expected discounted return $\mathbb{E}_{\pi, p_0} [\sum_{t=0}^{\infty} \gamma^{t-1} r_t(s_t, a_t, s_{t+1})]$
- Meta-learning
 - Given data from $\mathcal{T}_1, \dots, \mathcal{T}$, quickly solve new task $\mathcal{T}_{test}$

● Problem setting of Meta-Reinforcement Learning(Meta-RL)

- Setting 1: meta-learning with diverse goal(goal as a task)
 - $\mathcal{T}_i \triangleq \{\mathcal{S}, \mathcal{A}, p(s_0), p(s'|s, a), r(s, a, g), g_i\}$
- Setting 2: meta-learning with RL tasks(MDP as a task)
 - $\mathcal{T}_i \triangleq \{\mathcal{S}_i, \mathcal{A}_i, p_i(s_0), p_i(s'|s, a), r_i(s, a)\}$



Meta RL problem statement in CS-330(Finn)

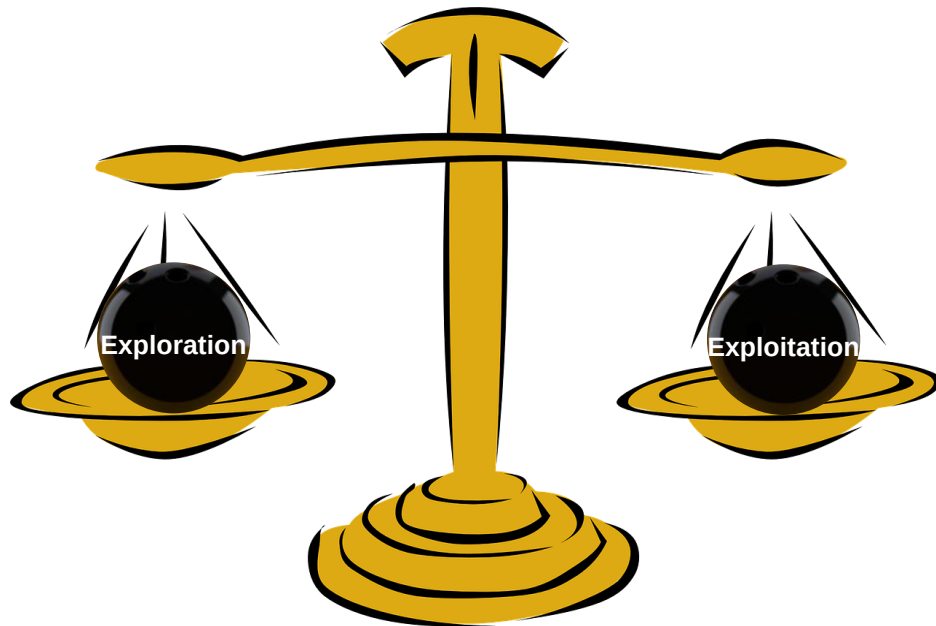
A Very Good Method for Bayes-Adaptive Deep RL via Meta-Learning(variBAD)

<https://arxiv.org/pdf/1910.08348.pdf>

Backgrounds

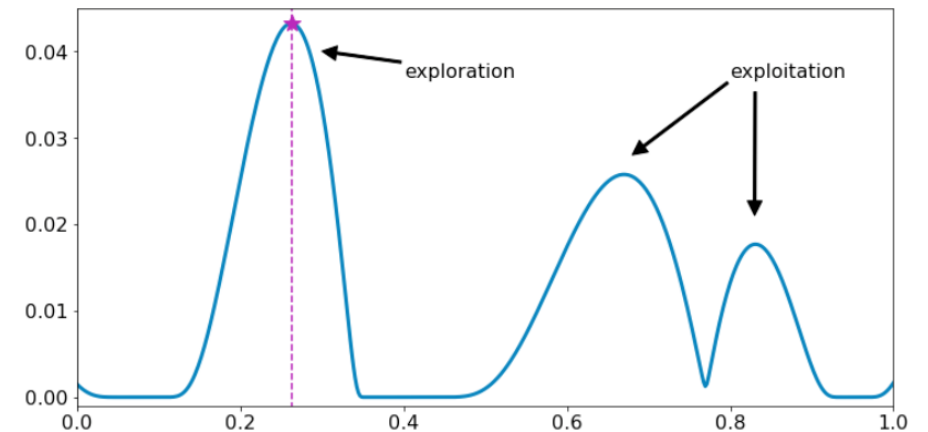
- **Exploration-exploitation dilemma**

- Agent learns the optimal policy through exploration and exploitation
 - The agent need to gather enough information to make the best overall decisions (exploration)
 - The agent need to make the best decision given current information (exploitation)



- **Bayesian Reinforcement Learning(Bayesian RL)**

- Bayesian methods for RL can be used to quantify uncertainty for action-selection, and provide a way to incorporate prior knowledge into the algorithms
- A Bayes-optimal policy is one that **optimally trades off exploration and exploitation**, and thus maximizes expected return during learning
 - Bayes-optimal policy is the solution of the reinforcement learning problem formalized as a **Bayes-Adaptive MDP(BAMDP)**



● Bayes Adaptive MDP(BAMDP)

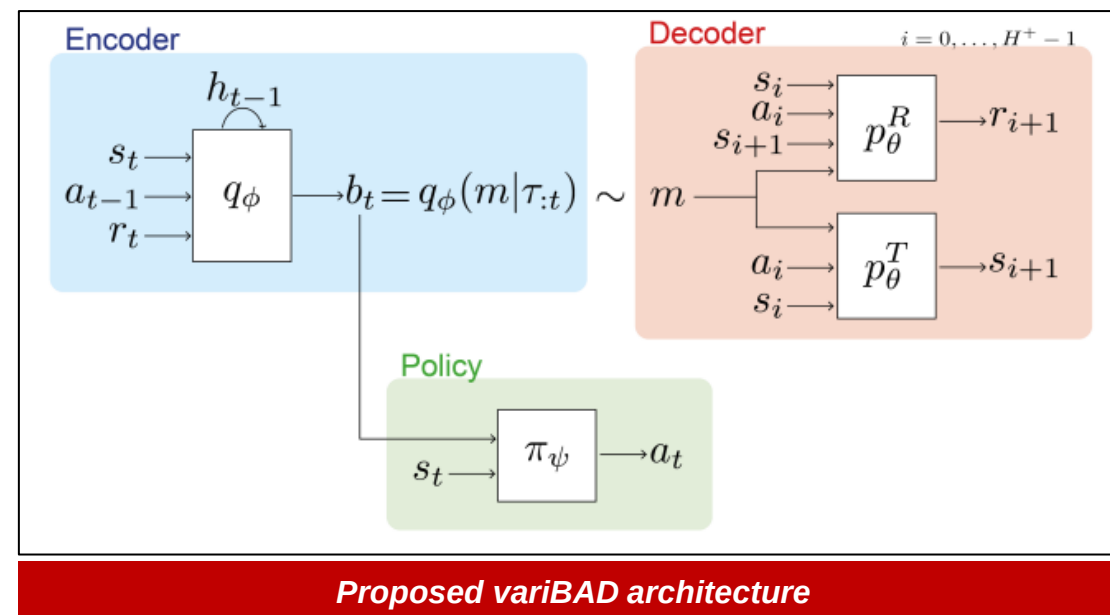
- A special case of PODMPs where transition and reward functions constitute the hidden state and the agent must maintain a belief over them.
- In the Bayesian formulation of RL, they assume that the transition and reward functions are distributed according to a prior $b_0 = p(R, T)$.
 - Agent does not have access to the true reward and transition function
- BAMDP $M^+ = (\mathcal{S}^+, \mathcal{A}, \mathcal{R}^+, T^+, T_0^+, \gamma, H^+)$
 - In this framework, the state is augmented with belief to incorporate the task uncertainty into decision-making
 - Hyper-states: $s_t^+ \in \mathcal{S}^+ = \mathcal{S} \times \mathcal{B}$, where $\mathcal{B}$ is the belief space
 - Transition on hyper-states: $T^+(s_{t+1}^+ | s_t^+, a_t, r_t) = T^+(s_{t+1}, b_{t+1} | s_t, a_t, r_t, b_t) = T^+(s_{t+1} | s_t, a_t, b_t) T^+(b_{t+1} | s_t, a_t, r_t, b_t, s_{t+1})$
 $= \mathbb{E}_{b_t}[T(s_{t+1} | s_t, a_t)] \delta(b_{t+1} = p(R, T | \tau_{t+1}))$
 - Reward function on hyper-states: $R^+(s_t^+, a_t, s_{t+1}^+) = R^+(s_t, b_t, a_t, s_{t+1}, b_{t+1}) = \mathbb{E}_{b_{t+1}}[R(s_t, a_t, s_{t+1})]$
 - Horizon: $H^+ = N \times H$
 - Objective of agent in the BAMDP: $\mathcal{J}^+(\pi) = \mathbb{E}_{b_0, T_0^+, T^+, \pi}[\sum_{t=0}^{H^+-1} \gamma^t R^+(r_{t+1} | s_t^+, a_t, s_{t+1}^+)]$
 - It is maximized by Bayes-optimal policy, which automatically trades off exploration and exploitation

● Overview of the variBAD

- The authors proposed variBAD to resolve some challenges of problem formulated as BAMDP, as follows
 - 1) Agent typically does not know the parameterization of the true reward and/or transition model
 - 2) The belief update (computing the posterior $p(R, T|\tau_{:t})$ is often intractable
 - 3) Even with the correct posterior, planning in belief space is typically intractable
- 1) They simultaneously meta-learned the reward and transition function with posterior over task embedding
 - $T' = p_{\theta}^T(s_{t+1}|s_t, a_t; m)$, $R' = p_{\theta}^R(r_{t+1}|s_t, a_t, s_{t+1}; m)$, $m \sim q_{\phi}(m|\tau_{:t})$ is a task embedding variable, q_{ϕ} is the encoder
- 2) The posterior over task embedding, $q_{\phi}(m|\tau_{:t})$ is trained when the decoder $p(\tau_{:t}|m)$ is updated through training objectives
 - Lower bound of model objective: $\text{ELBO}_t(\phi, \theta) = \mathbb{E}_{\rho} \left[E_{q_{\phi}(m|\tau_{:t})} [\log p_{\theta}(\tau_{:H}|m) + KL(q_{\phi}(m|\tau_{:t}) || p_{\theta}(m))] \right]$
 - Policy gradient objective: $\mathbb{E}_{p(M)} [\mathcal{J}(\psi, \phi)]$
- 3) Since their model is learned end-to-end with the inference framework, no planning is necessary at test time

● Bayes-adaptive deep RL via meta-learning

- The reward and transition functions that are unique to each MDP are learned with task description variable m_i given MDP $M_i \sim p(M)$
 - $R_i(r_{t+1}|s_t, a_t, s_{t+1}) \approx R(r_{t+1}|s_t, a_t, s_{t+1}; m_i)$
 - $T_i(s_{t+1}|s_t, a_t) \approx T(s_{t+1}|s_t, a_t; m_i)$
- Since the agent do not have access to the true task description or ID, they infer m_i given the agent's experience up to time step t collected in M_i
 - They infer the posterior distribution $p(m_i|\tau_{1:t}^i)$ over m_i given $\tau_{1:t}^i = (s_0, a_0, r_1, s_1, a_1, r_2, \dots, s_{t-1}, a_{t-1}, r_t, s_t)$ from now on, the subscript i is dropped for ease of notation)
- Given the reformulation, it is now sufficient to reason about the embedding m , instead of the transition and reward dynamics
 - The authors suggested that this is significantly useful, where the reward and transition function can consist of millions of params, but the embedding m can be small



● Model learning objective

- They learned a model of the environment p_θ , so that even if they don't know the model, they could allow fast inference at runtime at using follow objective

- $\mathbb{E}_{\rho(M, \tau_{:H^+})} [\log p_\theta(\tau_{:H^+} | a_{:H^+-1})]$

- Instead of optimizing above objective, which is intractable, they optimized a tractable lower bound

- $$\begin{aligned} \mathbb{E}_{\rho(M, \tau_{:H^+})} [\log p_\theta(\tau_{:H})] &= E_\rho \left[\log \mathbb{E}_{q_\phi(m|\tau_{:t})} \left[\frac{p_\theta(\tau_{:H}, m)}{q_\phi(m|\tau_{:t})} \right] \right] \geq \mathbb{E}_{\rho, q_\phi(m|\tau_{:t})} \left[\log \frac{p_\theta(\tau_{:H}, m)}{q_\phi(m|\tau_{:t})} \right] \\ &= \mathbb{E}_\rho \left[E_{q_\phi(m|\tau_{:t})} [\log p_\theta(\tau_{:H} | m) + KL(q_\phi(m|\tau_{:t}) || p_\theta(m))] \right] = ELBO_t \end{aligned}$$

● Overall objective

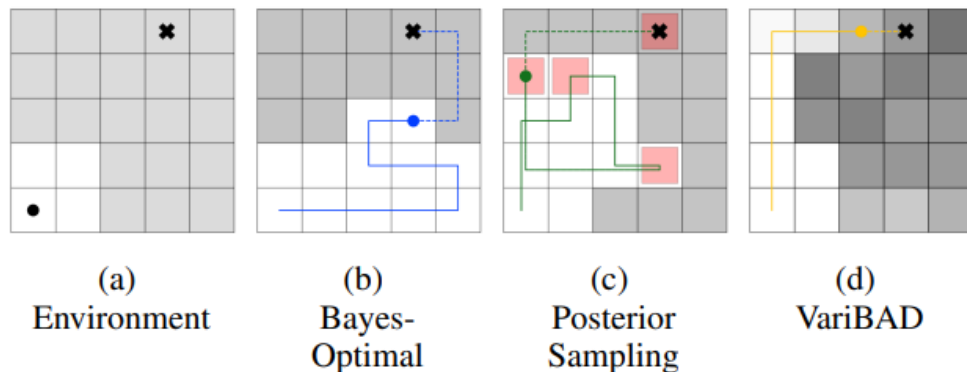
- The encoder $q_\phi(m|\tau_{:t})$, parameterized by ϕ
- An approximate transition function $T' = p_\theta^T(s_{t+1}|s_t, a_t; m)$ and an approximate reward function $R' = p_\theta^R(r_{t+1}|s_t, a_t, s_{t+1}; m)$ which are jointly parameterized by θ
- A policy $\pi_\psi(a_t|s_t, q_\phi(m|\tau_{:t}))$ parameterized by ψ and dependent on ϕ
 - $\mathcal{L}(\phi, \theta, \psi) = \mathbb{E}_{p(M)} [\mathcal{J}(\psi, \phi) + \lambda \sum_{t=0}^{H^+} ELBO_t(\phi, \theta)]$

Experiment Results

Experiment environment

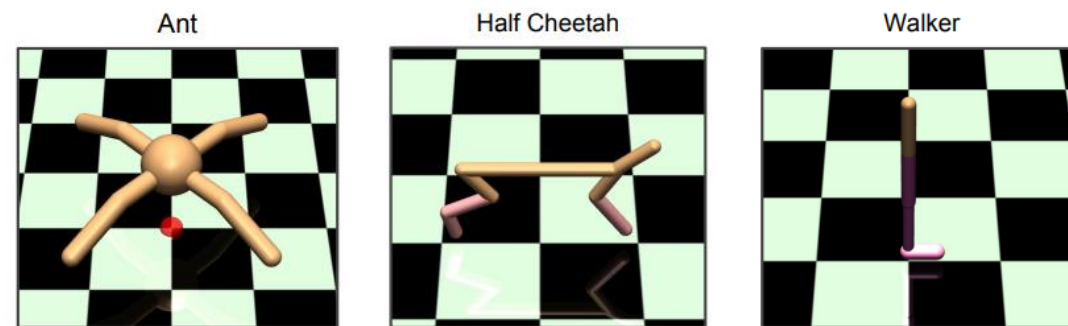
● Environment

- Didactic grid world
 - The goal is **unobserved** by the agent, inducing task uncertainty and necessitating exploration.
 - The goal can be anywhere except around the starting cell, which is at the bottom left.
- MDP parameters
 - Action: up, right, down, left, stay
 - Reward: -0.1 on non-goal cells, +1 on the goal cell
 - Horizon: 15



Different exploration strategies on didactic grid world

- MuJoCo for meta-learning
 - Well-known environment for continuous control
 - Ant-Dir: move forward and backward (2 tasks)
 - Half-Cheetah-Dir: move forward and backward (2 tasks)
 - Half-Cheetah-Vel: Achieve a target velocity running forward (100 train tasks, 30 test tasks)
 - Walker: Agent initialized with some system dynamics params randomized and move forward (40 train tasks, 10 test tasks)



Continuous control tasks: the ant, half cheetah, and walker robots

Experiment results (I)

● Comparison with the previous method on didactic grid world

- Experiments show the superior performance of variBAD compared to previous methods in didactic grid-world environment with few episodes

- variBAD



- RL²



- Optimal



- Posterior Sampling

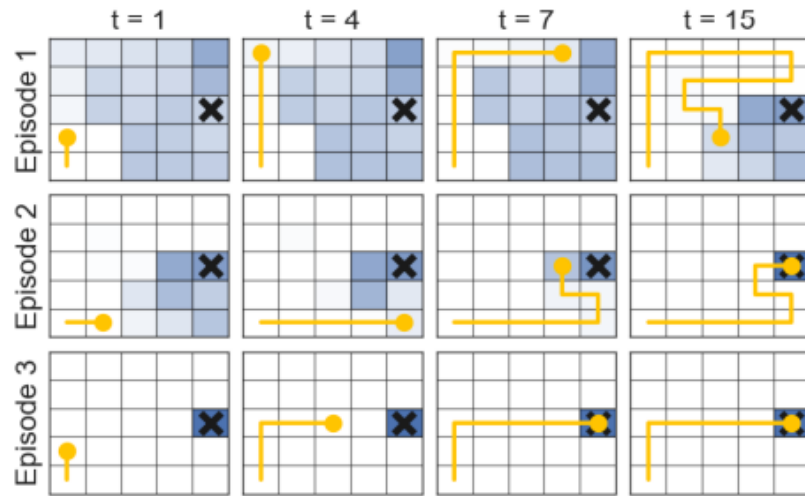


- Bayes-Optimal

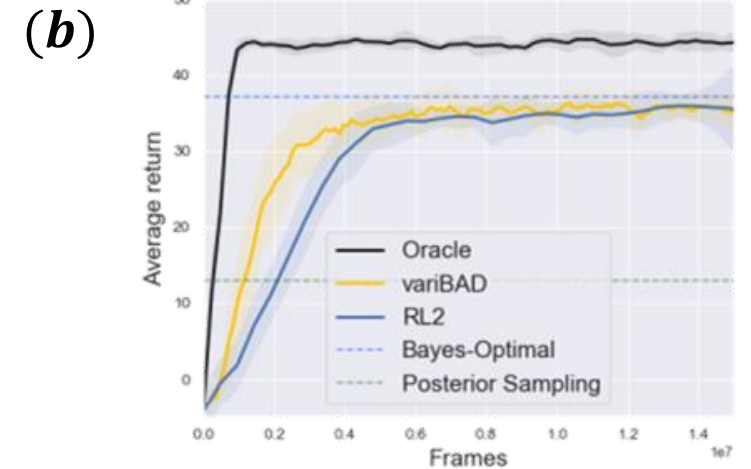
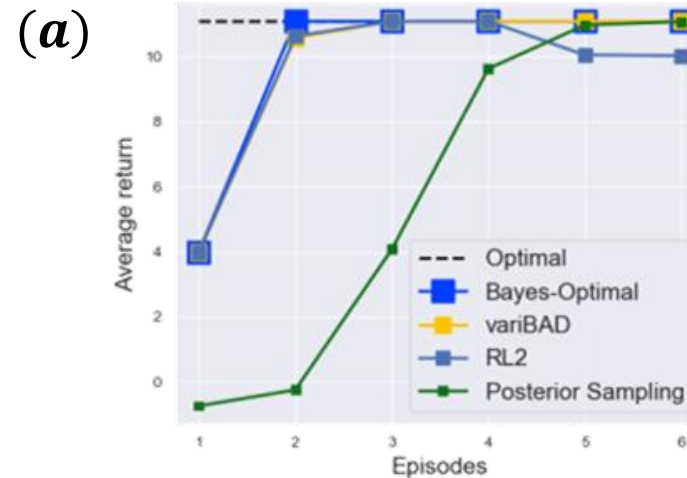


- Behavior of variBAD closely matched that of the Bayes-optimal policy

- The variBAD could predicts no reward for cells it has visited, and explores the remaining cells until it finds the goal



Behavior of variBAD in the grid-world



Results for the grid-world (a) Average return (b) running curves per frame (20 seeds)

Experiment results (II)

● Comparison with the previous method on MuJoCo

- Experiments show the superior performance of variBAD compared to previous methods in MuJoCo environment (Ant-Dir, Cheetah-Dir, Cheetah-Vel, Walker) with few episodes

- variBAD



- PEARL



- Oracle(PPO)



- EMAML



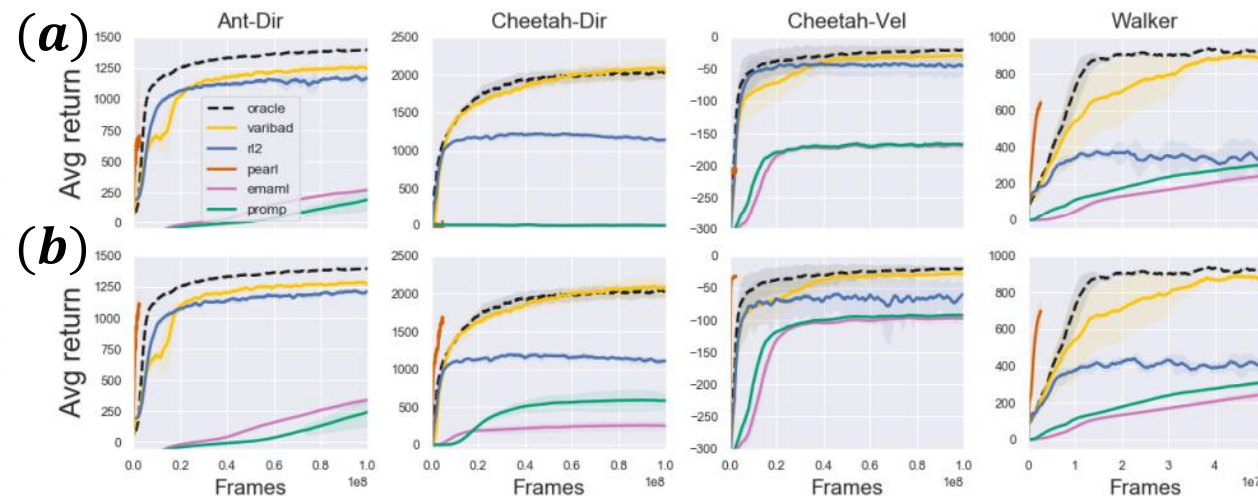
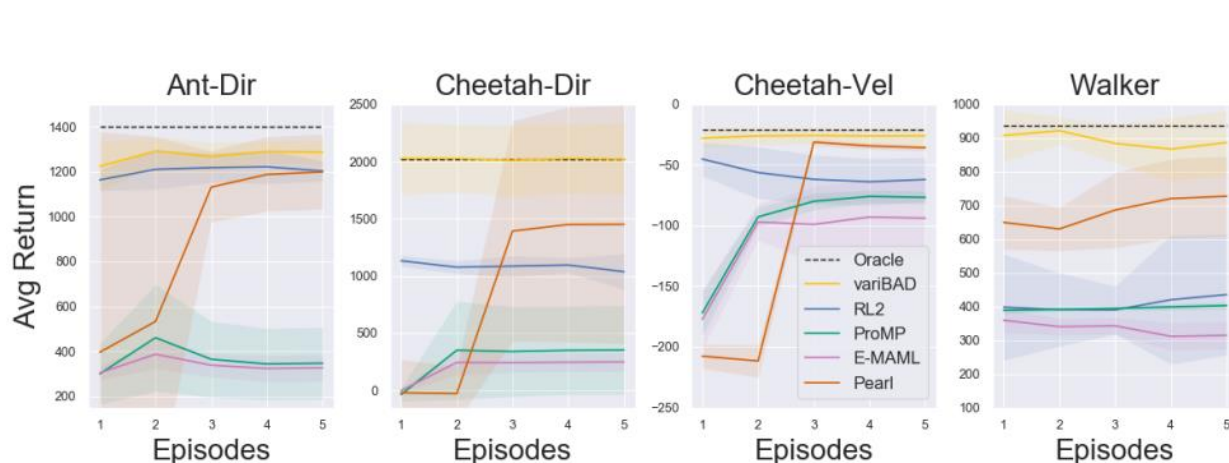
- RL²



- PROMP



- For all MuJoCo environments, they trained variBAD with a reward decoder only (even for Walker, where the dynamics change, we found that this has superior performance)



Average test performance for the first 5 rollouts of MuJoCo (5 seeds)

Learning curves for the MuJoCo at the (a) first rollout (b) N-th rollout

Thank you!

Q&A